

B E E F T E C H

Livestock traceability and farm data capture for field teams



BeefTech Mobile Data Collection Platform

User Manual

Field agents

Site managers

Administrators

IT & installers

App version
1.0 (Android 7.0 and later)

Manual version
1.0

Date
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CHAPTER 1

Introduction

What the app does

The BeefTech app lets field teams record livestock and farm information on an Android phone, even where there is no signal. Records are kept safely on the phone and sent to the BeefTech server as soon as the phone is online.

With the app you can:

- **Register calves** with their ear tag, breed, birth details, condition and parents.
- **Register farmers** (clients), their address, farm details and GPS location.
- **Trace animals**: movements, supplier and purchase details, location and feed, treatments, costs, weighings and deaths.
- **Link animals to farmers** so each farmer's herd is known.
- **Manage the operation** (managers and admins): dashboards, reports, record review, team, phones, sites and system settings.

Roles and what each one sees

Every person signs in with their own username and 5-digit PIN. What they can do depends on their role. Higher roles can do everything the lower roles can.

FIELD AGENT**Worker**

Captures calves, farmers and traceability records. Sees their own sync status and activity.

SITE MANAGER**Manager**

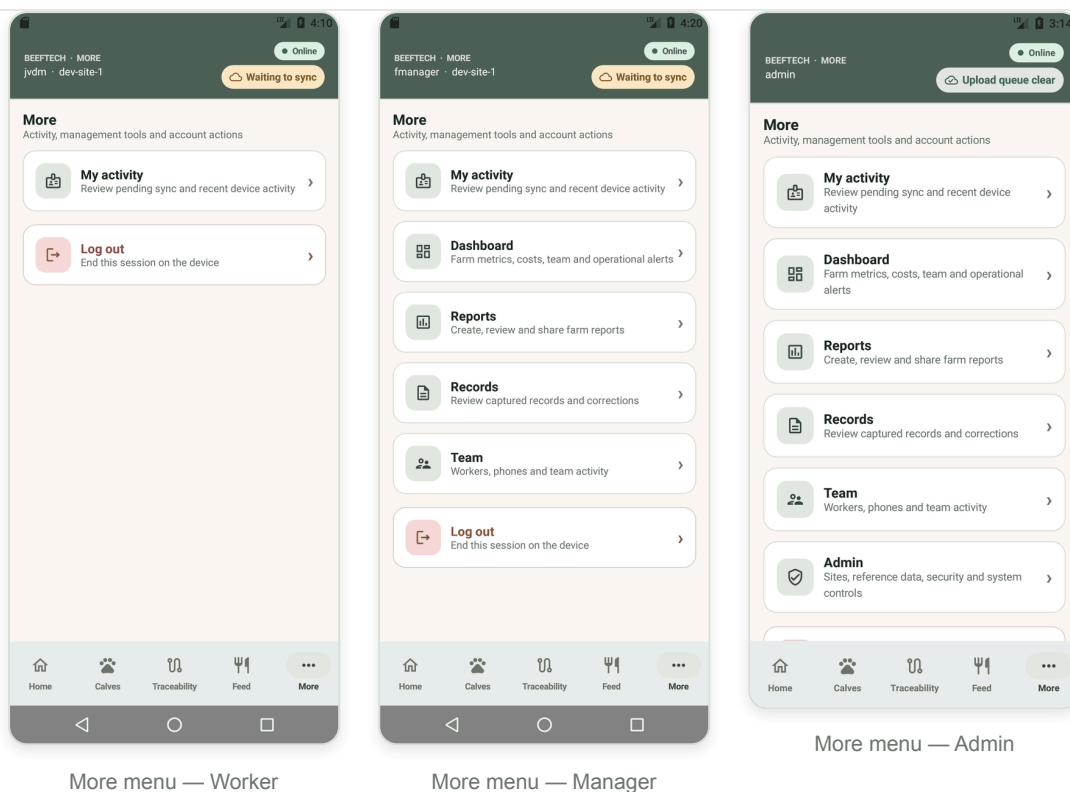
Everything a worker can do, plus Dashboard, Reports, Records and Team for **their own site**.

SYSTEM OWNER**Admin**

Everything a manager can do across **all sites**, plus the Admin area: sites, phones, security, reference data and sync policy.

NOTE

On the phone the app may still be labelled **demoapp** in the app drawer. It is the BeefTech app.



More menu — Worker

More menu — Manager

More menu — Admin

Finding your way around

After signing in, the bar at the bottom of the screen has five tabs. They are the same for every role:

Tab	What it is for
Home	Welcome, quick actions (Register calf, Traceability, Feed; Dashboard and Reports for managers) and the Sync card.
Calves	Register a calf or look up calves registered on this phone.
Traceability	Farmers, animal assignment and every per-animal record: movements, supplier, location & feed, treatments, costs and mortality.
Feed	Feed crib readings: pick a crib, enter a reading code and the A.D.I. See Feed tab .
More	My activity and Log out. Managers and admins also find their management tools here.

The green header at the top of every screen shows where you are (for example **BEEFTECH · CALVES**), who is signed in and on which site, whether the phone is **Online** or **Offline**, and the upload status pill (see [Reading the sync status](#)).

How to read this manual

- Buttons and labels you tap appear like this: **Save calf**.
- Fields marked with * in the app are required. The app will not save until they are filled in.
- The screenshots were taken with sample data on a test system. Your names, sites and numbers will differ.



Getting started

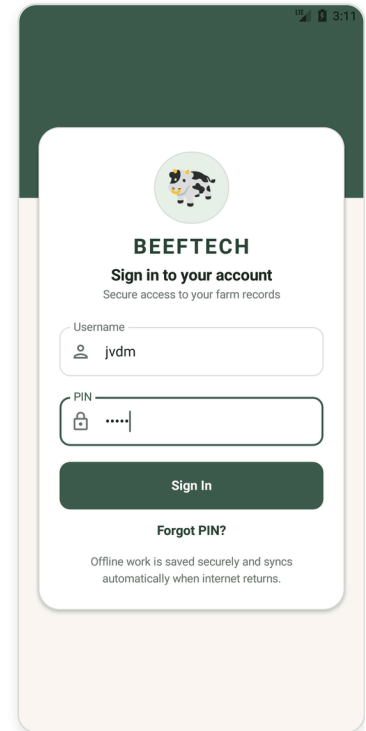
Signing in

- 1 Open the BeefTech app.
- 2 Type your **Username**.
- 3 Type your 5-digit **PIN**. The **Sign In** button only becomes active once both are filled in.
- 4 Tap **Sign In**.

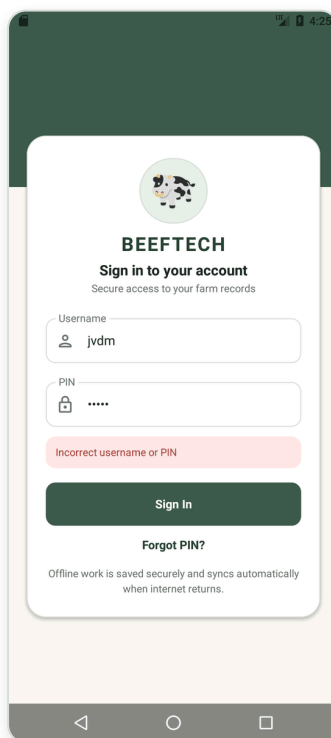
FIRST TIME ON A PHONE

The very first time you sign in on a phone it must be **online**. Your account is then linked to that phone, and from then on you can also sign in without signal.

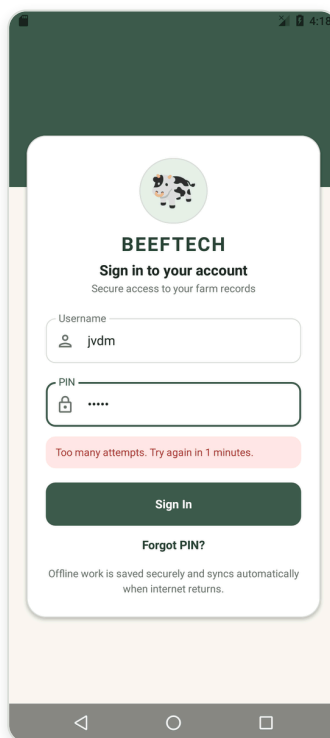
If you type the wrong PIN five times in a row, the phone blocks sign-in for **1 minute**. If this happens while online, the server also locks your account for **5 minutes**. A manager can unlock it sooner from [Team](#).



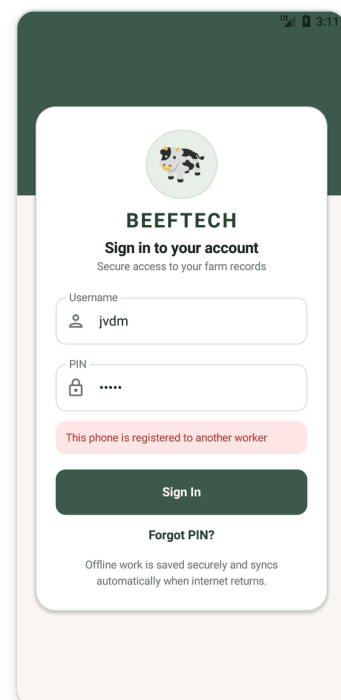
Sign-in screen



Wrong username or PIN



After five wrong PINs

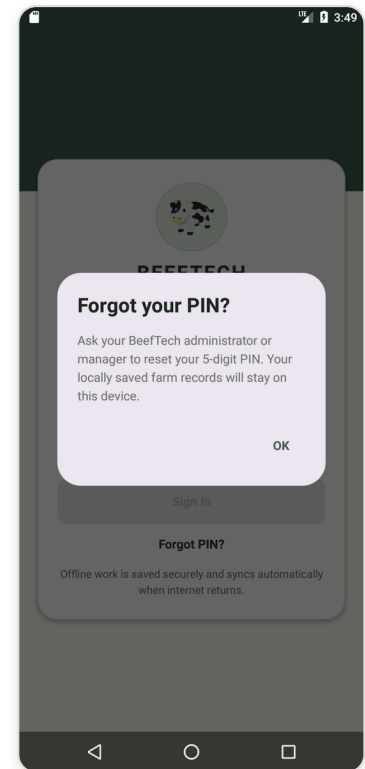


Account linked to another phone

Forgot your PIN

Tap **Forgot PIN?** on the sign-in screen. You cannot reset a PIN yourself. Ask your manager or administrator to give you a new one (see [Team → Reset PIN](#)).

Records already saved on the phone are **not** lost when your PIN is reset.

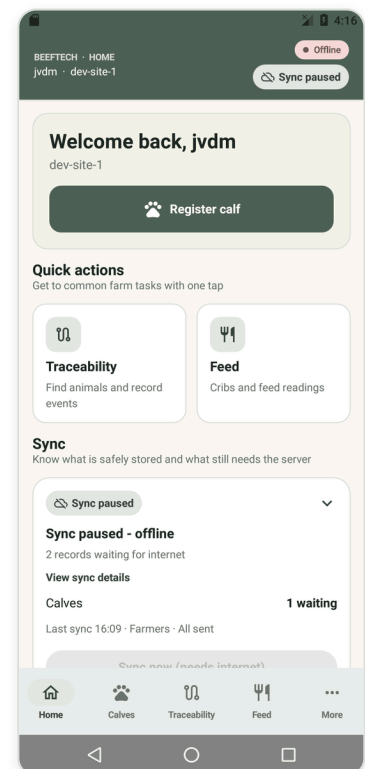


Forgot PIN

Signing in without signal

Once you have signed in online on a phone, you can sign in on that phone with no signal. The app checks your PIN against a secure copy kept on the phone.

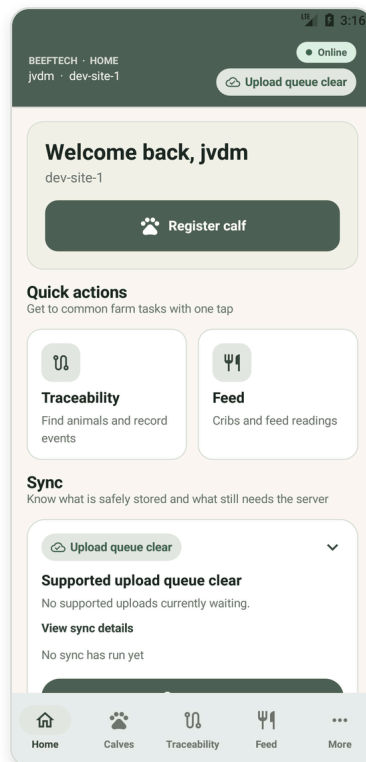
- An offline session lasts up to **7 days**. After that, connect and sign in online again.
- Everything you capture offline is saved on the phone and sent later.
- The header shows **Offline** and **Sync paused** while there is no connection.



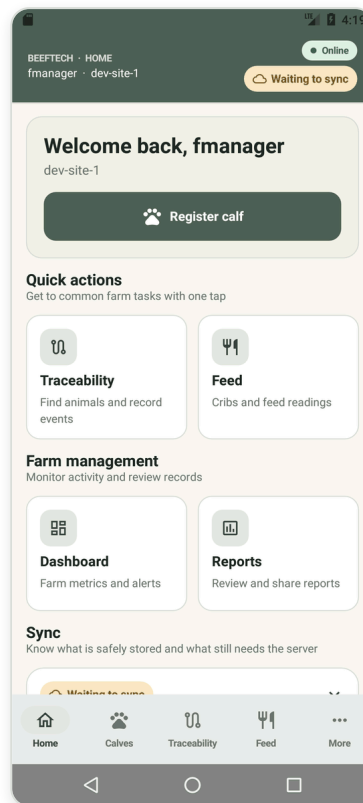
Signed in while offline



The Home screen



Home — Worker



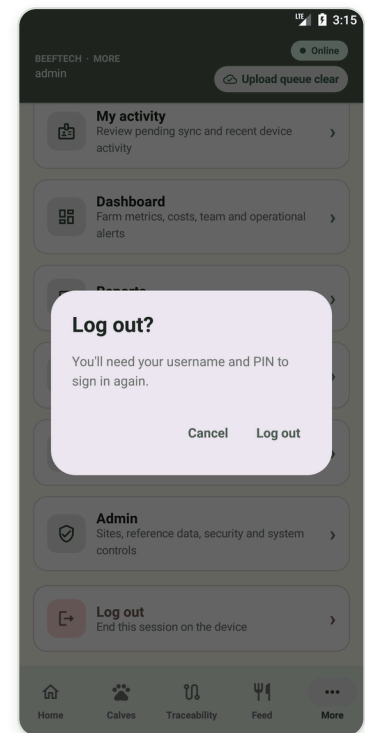
Home — Manager, with Farm management tiles

- **Welcome back** shows your username and site.
- **Register calf** starts a calf registration straight away.
- **Quick actions** open Traceability and Feed. Managers and admins also get **Farm management** tiles for Dashboard and Reports.
- The **Sync** card tells you what is safely stored on the server and what is still waiting (see [Chapter 3](#)).

Logging out

Go to **More** → **Log out** and confirm with **Log out**. You will need your username and PIN to sign in again.

Log out when you hand the phone to someone else, so their records are captured under their own name.



Log out confirmation



Working offline and syncing

How saving works

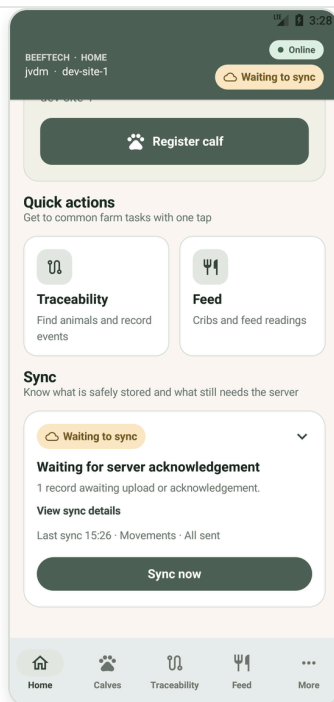
Every record is saved on the phone first, then sent to the server. You never need signal to capture data, but you do need signal for the data to reach the office.

- 1 You tap **Save**. The record is stored, encrypted, on the phone and marked **PENDING**.
- 2 When the phone has a connection, the app uploads waiting records in the background.
- 3 When the server confirms it received a record, the record becomes **SYNCED**.
- 4 If the server refuses a record (for example the ear tag is already registered), it is marked **NEEDS ATTENTION** and the reason is shown on the record.

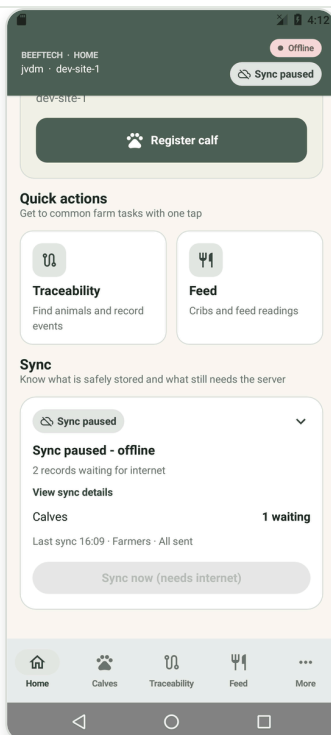
Reading the sync status

The pill in the header and the Sync card on Home always show the current state:

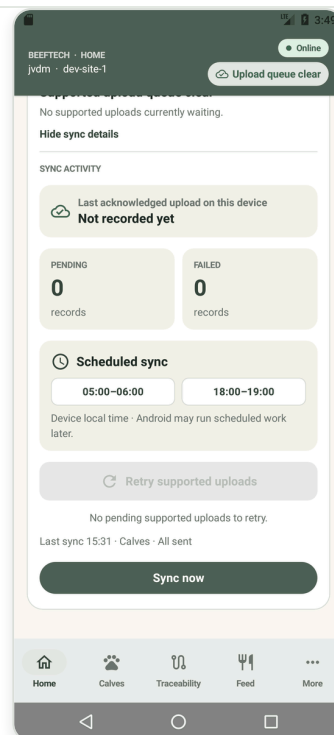
Status	Meaning
Upload queue clear	Everything captured on this phone has been sent.
Waiting to sync	Records are queued and will be sent shortly, or the server has not yet confirmed them.
Sync paused	The phone is offline. The card shows how many records are waiting for internet.
Needs attention	One or more records could not be sent. Open the record to see why and use Retry sync once fixed.



Online, waiting for the server



Offline: records waiting per module



View sync details

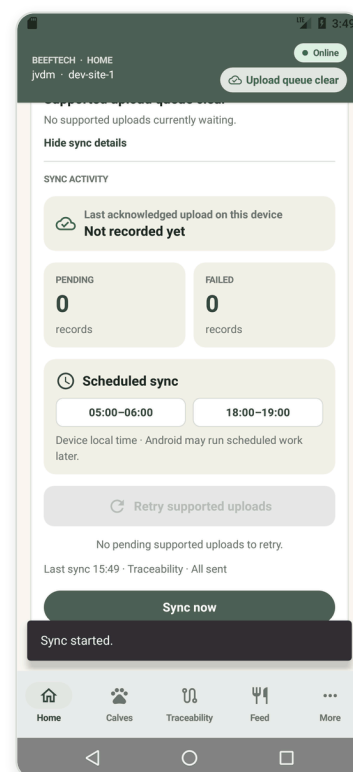
Tap **View sync details** on the Sync card to see the numbers of **PENDING** and **FAILED** records, the time of the last upload the server confirmed, and the scheduled sync windows.

Sync now and scheduled sync

- **Automatic:** waiting records upload by themselves as soon as the phone has a connection.
- **Scheduled:** the app also runs a full sync every day between **05:00–06:00** and **18:00–19:00** (phone time). Android may run it a little later to save battery.
- **Sync now:** tap **Sync now** on the Home Sync card to send everything immediately. You will see *Sync started*. While offline the button reads **Sync now (needs internet)**.
- **Retry supported uploads** (in sync details) tries again for records that failed.

GOOD HABIT

Before leaving a farm or at the end of the day, find signal and tap **Sync now**. Wait until the pill shows **Upload queue clear**.

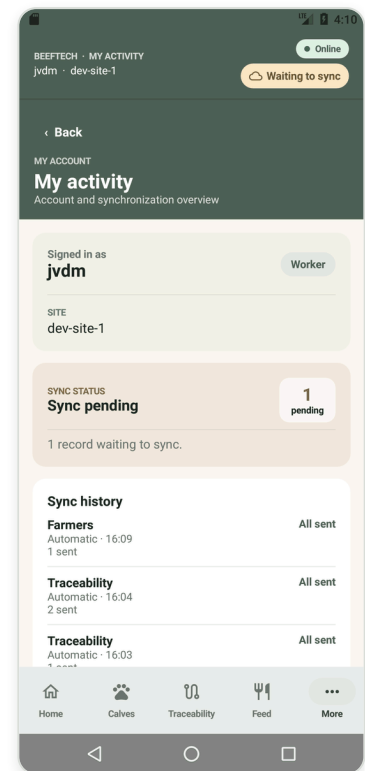


Sync started

Sync history

More → **My activity** shows your account, site, how many records are still waiting, and a **Sync history** of the last 30 days.

Each entry shows the module (Calves, Farmers, Traceability, ...), what started it — *Morning sync*, *Evening sync*, *Sync now* or *Automatic* — the time, and a summary such as *3 sent*, *1 waiting* with **All sent**, **Partly sent**, **Failed** or **Offline**.



My activity

The 7-day rule

To protect the data, the app does not let records sit unsynced on a phone forever. Counting from the oldest unsynced record:

Day (default)	What you see	What to do
Day 2	Data Not Synced	Connect to a network and sync your records.
Day 4	Critical Sync Warning	Connect and sync as soon as possible.
Day 6	Final Sync Warning	Connect and sync today .
Day 7	Account locked	The unsynced records are deleted from the phone and the account is locked until an administrator restores access.

IMPORTANT

On day 7, records that were never sent are **permanently deleted** from the phone. Do not ignore the warning banners. The administrator can move the warning days (see [Sync policy](#)) but day 7 is fixed.

Field agent guide

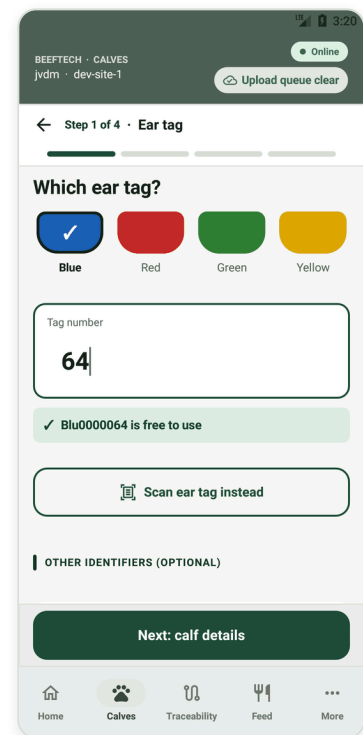
This chapter covers the day-to-day capture work: calves, farmers and the traceability records for each animal.

Registering a calf

Open the **Calves** tab and tap **Register a calf**, or tap **Register calf** on Home. Registration has four steps; the bar at the top shows where you are, and the back arrow returns to the previous step without losing what you typed.

Step 1 — Ear tag

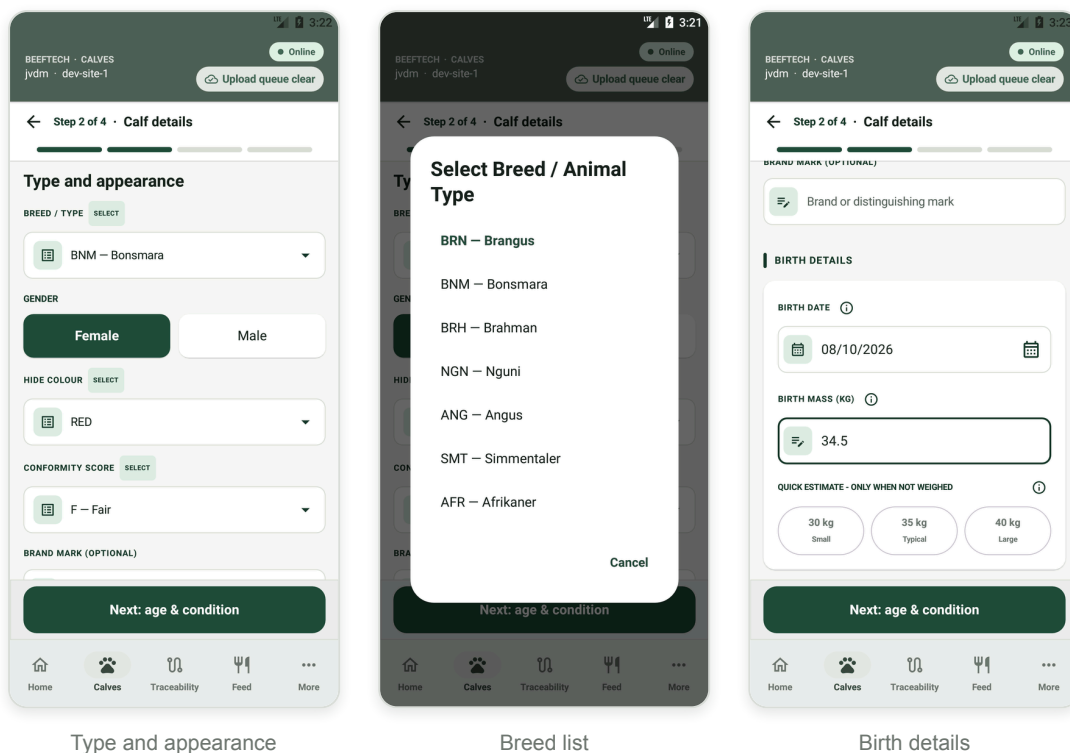
- 1 Choose the tag colour: **Blue**, **Red**, **Green** or **Yellow**.
- 2 Type the number printed on the tag, for example **64**. The app builds the full tag ID — Blue 64 becomes **Blu0000064**.
- 3 Watch the line below the field: “*Blu0000064 is free to use*” means you can continue; if the tag is already registered the app tells you.
- 4 Or tap **Scan ear tag instead** to read the tag with the camera (see [Scanning an ear tag](#)).
- 5 Optionally fill in **Other identifiers**: old tag number and reference number.
- 6 Tap **Next: calf details**.



Step 1 — Ear tag



Step 2 — Calf details

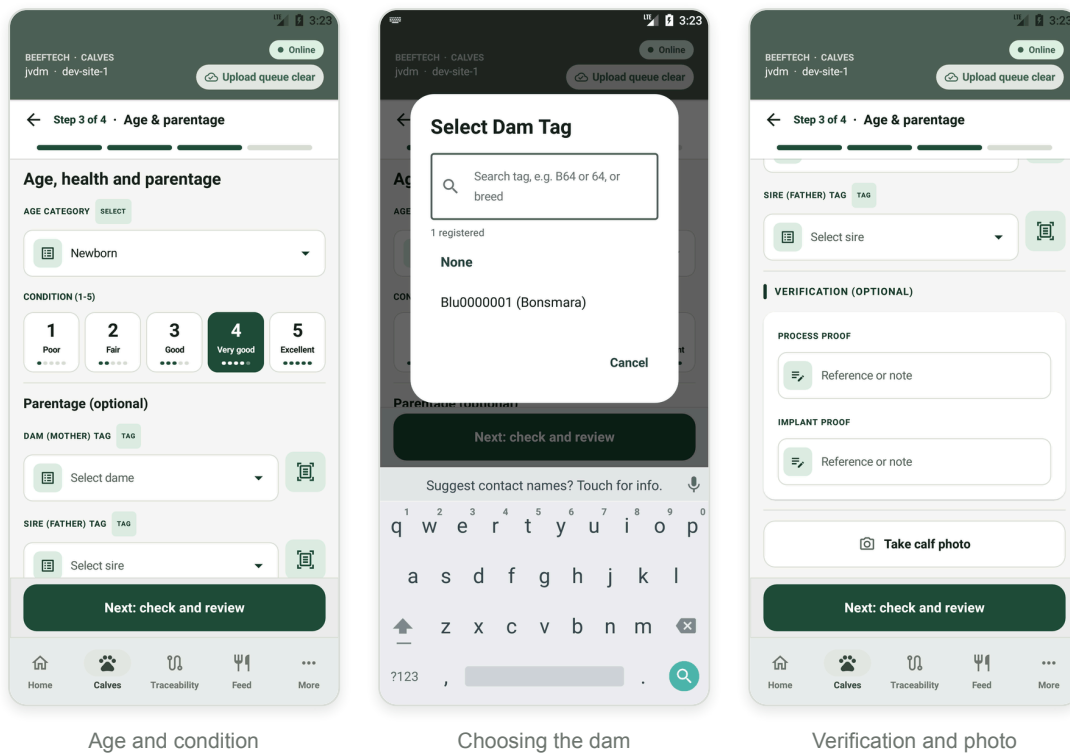


- **Breed / type:** BRN Brangus, BNM Bonsmara, BRH Brahman, NGN Nguni, ANG Angus, SMT Simmentaler or AFR Afrikaner.
- **Gender:** Female or Male.
- **Hide colour** and **conformity score:** choose from the lists.
- **Brand mark** (optional): any brand or distinguishing mark.
- **Birth date:** tap the calendar icon and pick the date.
- **Birth mass (kg):** the weighed mass. If the calf was *not* weighed, tap a quick estimate — **30 kg** small, **35 kg** typical or **40 kg** large.

Tap **Next: age & condition**.



Step 3 — Age, health and parentage

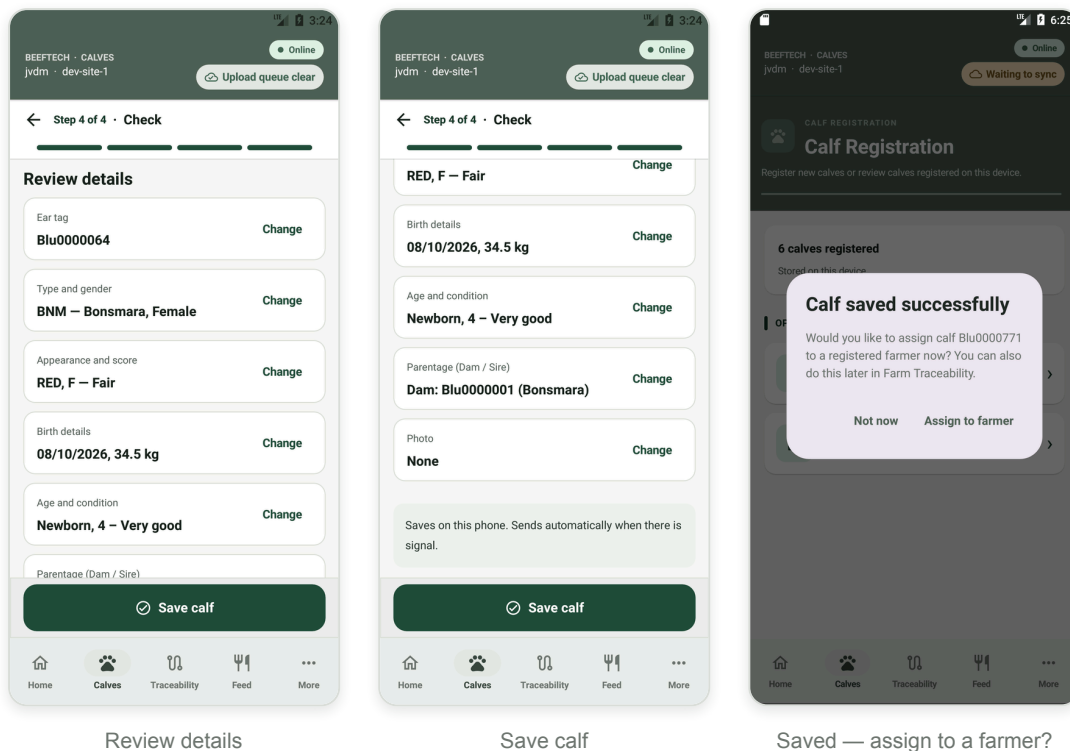


- **Age category:** Newborn, < 1 week, 1–2 weeks or > 2 weeks.
- **Condition (1–5):** 1 Poor, 2 Fair, 3 Good (default), 4 Very good, 5 Excellent.
- **Parentage (optional):** tap the dam (mother) or sire (father) field and search for a registered animal by tag (e.g. B64) or breed, or tap the scan button beside the field to scan the parent's tag.
- **Verification (optional):** process proof and implant proof references.
- **Take calf photo** (optional) opens the camera.

Tap **Next: check and review** .



Step 4 — Review and save



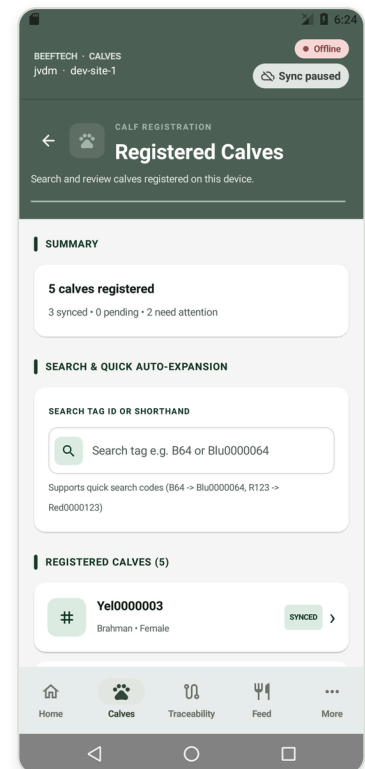
- 1 Check every section. Tap **Change** next to anything that is wrong to jump back to that step.
- 2 Tap **Save calf**. The calf is saved on the phone and will be sent when there is signal.
- 3 The app asks whether to assign the calf to a farmer now. Tap **Assign to farmer** (see [Assigning animals](#)) or **Not now**.

Registered calves

On the **Calves** tab tap **View registered calves**. The summary shows how many calves are registered on this phone and how many are synced, pending or need attention.

Search shortcuts: type the first letter of the colour and the number — **B64** finds **Blu00000064**, **R123** finds **Red00000123**. **G** is Green and **Y** is Yellow.

Each calf shows its status: **SYNCED**, **PENDING** or **NEEDS ATTENTION**. Tap a calf to see all its details (read-only).



Registered calves

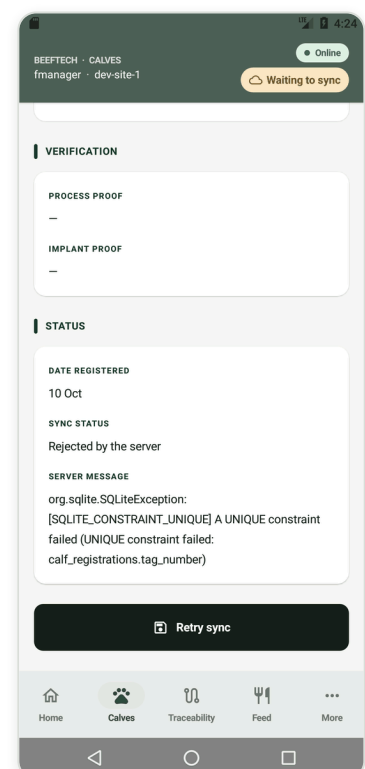
When a calf needs attention

Open the calf. At the bottom, **Sync status** reads *Rejected by the server* and **Server message** gives the reason. The most common reason is that the **ear tag is already registered** on the server.

- If the tag was mistyped, register the calf again with the correct tag, and ask your manager to void the wrong record.
- If the problem was temporary, tap **Retry sync**.

NOTE

The server message can be technical. Text containing *UNIQUE constraint ... tag_number* means the tag number already exists.



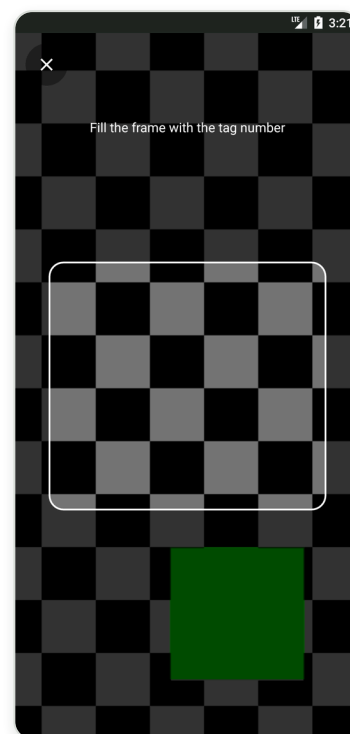
Rejected calf with Retry sync

Scanning an ear tag

Wherever you see **Scan ear tag** (calf step 1, the dam and sire fields, and Find Animal):

- 1 Allow camera access the first time the app asks.
- 2 Hold the phone so the **tag number fills the frame**. Use the torch button in poor light.
- 3 The app reads the number and the tag colour. If it cannot tell the colour, it asks you to tap it.
- 4 Tap **Use this tag**, or **Rescan** to try again.

If camera access was refused, choose **Enter manually** and type the number instead.



Scanner (emulator camera shown)

Registering a farmer

Go to **Traceability** → **Farmer Registration**. The list shows farmers registered on this phone with their status: **Pending Sync**, **Processing** or **Registered**. Tap **Register farmer** to add one. There are three screens.

1. Client Details

Identity profile

Email, VAT and roles

Saved farmer in the list

Field	Rules
Client code	Required and must be unique. The app warns "Client Code already exists!".
Co-Reg / ID no., name of organisation	Company registration or ID number, and the business or farm name.
Contact person	Optional. The person the sales rep should call.
Contact number	Optional. Only + , digits and spaces, with 9 to 15 digits, e.g. +27 82 555 0101 .
Email, VAT number	Optional.
Roles	Tick all that apply: Agent, Buyer, Client, Location, Feedlot, Owner, Supplier, Transporter.

Tap **Continue** , or **Discard** to abandon the registration.

2. Address & Location

The screenshots illustrate the 'Address & Location' registration process in the BEEFTECH app. The first screen shows a list of provinces. The second screen shows fields for Country, Land Ownership, Herd Capacity, Farm Size, Head Count, and Primary Breed. The third screen shows fields for Interest Status, FA Code, and GLN Number.

Province list

Land and herd

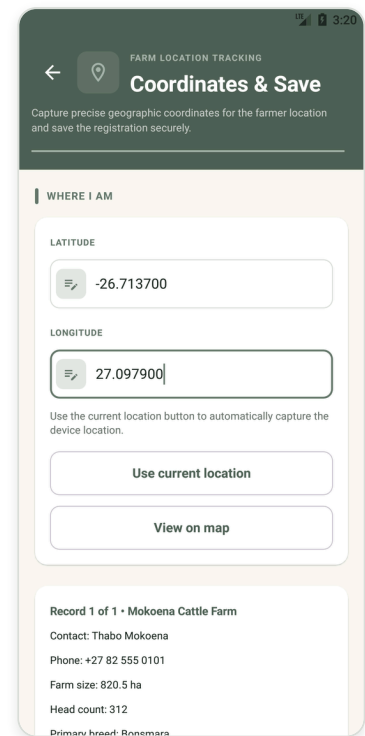
Interest, FA code and GLN

- **Street address, street code, postal code, postal address.**
- **Province** (the nine South African provinces) and **Country** (South Africa or a neighbouring country).
- **Land ownership:** Owned, Leased, Communal, State-owned or Trust.
- **Herd capacity** and **Head count:** whole numbers from 0 to 100 000 (optional).
- **Farm size (ha):** up to 2 decimals (optional). **Primary breed** (optional).
- **Interest status:** Interested, Follow-up needed, Not interested or Already a client (optional).
- **FA Code (RMIS):** pick a saved code or type it. **GLN number:** the 13-digit Global Location Number, or leave blank.

3. Coordinates & Save

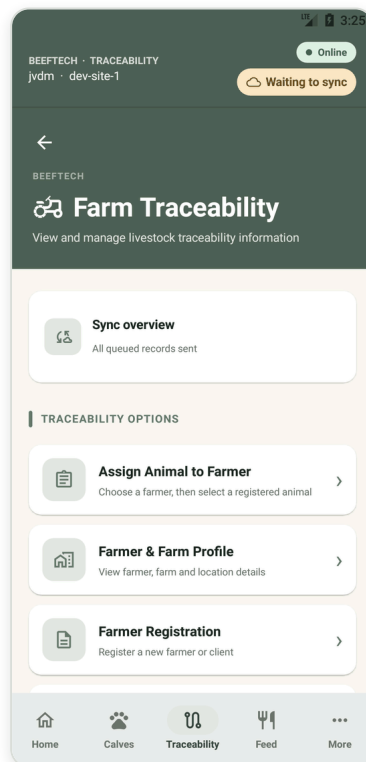
- 1 Stand at the farm and tap **Use current location**. Allow location access if asked. Or type the latitude (–90 to 90) and longitude (–180 to 180).
- 2 Tap **View on map** to check the position in your map app.
- 3 Check the summary card and tap **Save Location Data**.

The farmer is saved on the phone as **Pending Sync** and becomes **Registered** once the server has it. If the site has a sales rep, they are emailed about the new farmer.

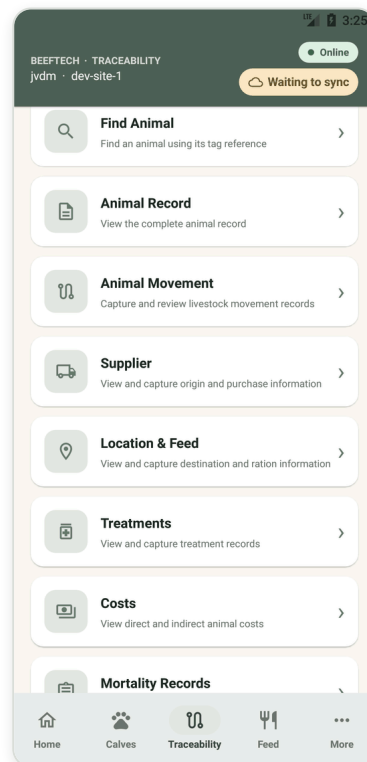


Coordinates & Save

Farm Traceability



Sync overview and farmer options



Per-animal options

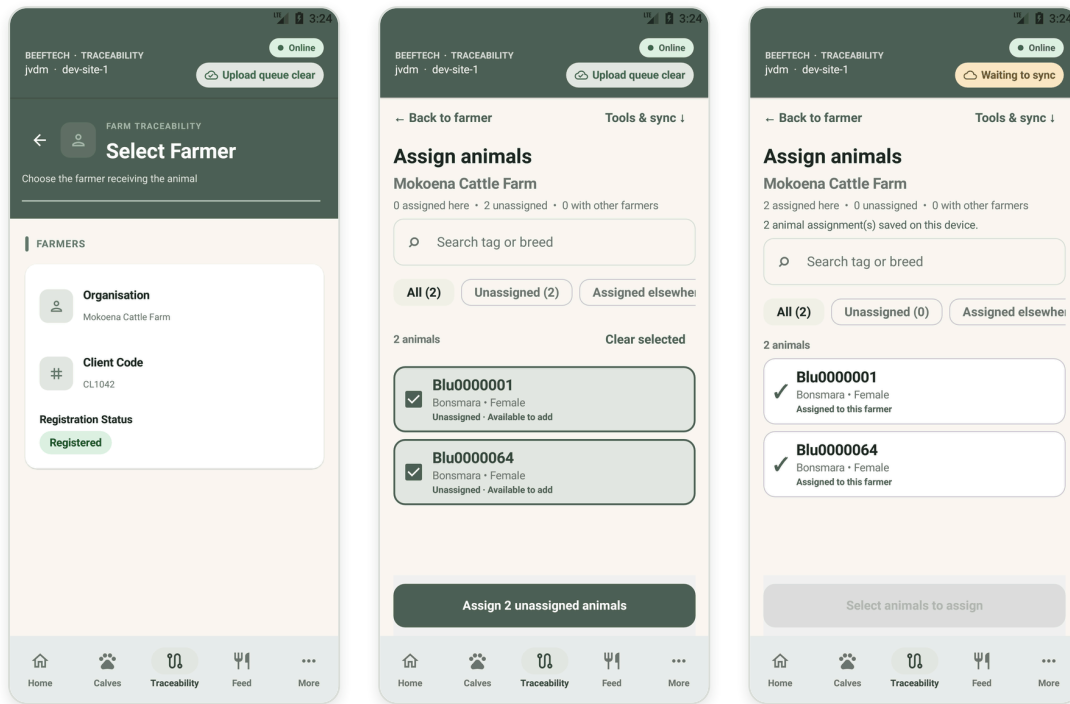
The **Traceability** tab starts with a **Sync overview** (and the day 2/4/6 warning banners when they apply). Below are the options:



- **Farmer options:** Assign Animal to Farmer, Farmer & Farm Profile, Farmer Registration.
- **Animal options:** Find Animal, Animal Record, Animal Movement, Supplier, Location & Feed, Treatments, Costs and Mortality Records.

Every animal option first opens **Find Animal** so you can pick the animal, then opens the form for that animal. The animal's tag is shown at the top of each form (e.g. *ANIMAL GRN0000007*).

Assigning animals to a farmer



1. Select the farmer

2. Tick the animals

3. Assigned

- 1 Open **Assign Animal to Farmer** and tap the farmer receiving the animals.
- 2 Use the search box or the filters **Unassigned**, **Assigned elsewhere** and **Assigned here** to find animals, and tick them.
- 3 Tap **Assign N unassigned animals**. The assignments are saved on the phone and synced.
- 4 To take an animal over from another farmer, select it under **Assigned elsewhere**. The app asks *Transfer {tag}?* and ends the old assignment when you confirm.

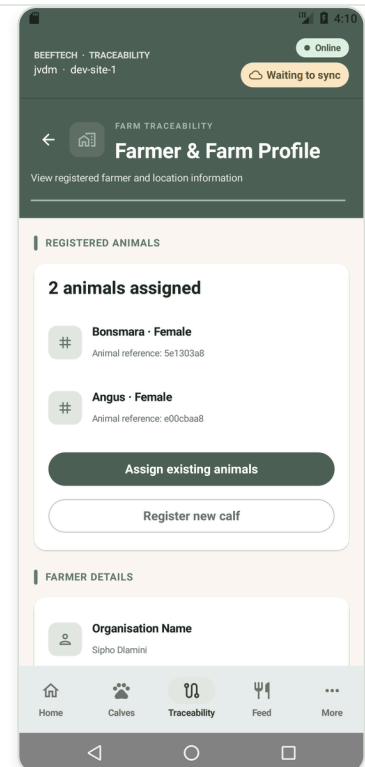
Tools & sync

Tap **Tools & sync** at the top of the screen for:

- **Download farm animals for offline use** — copies the site's animals to the phone before you go out of signal.
- **Check pending assignment issues** — reports assignments that have not reached the server.

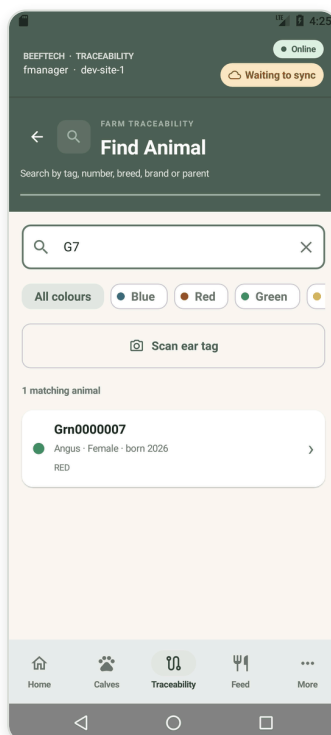
Farmer & Farm Profile

Shows a farmer's details, farm and the animals assigned to them, with buttons to **Assign existing animals** or **Register new calf**.

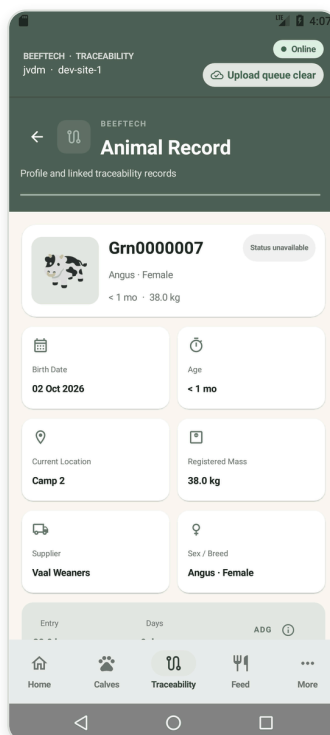


Farmer & Farm Profile

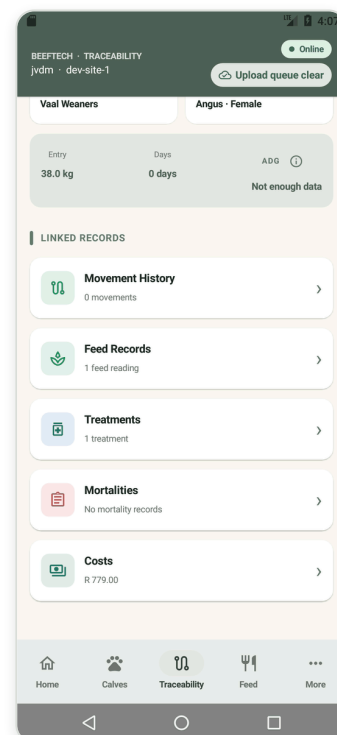
Find Animal and Animal Record



Find Animal with shortcut search



Animal Record



Linked records

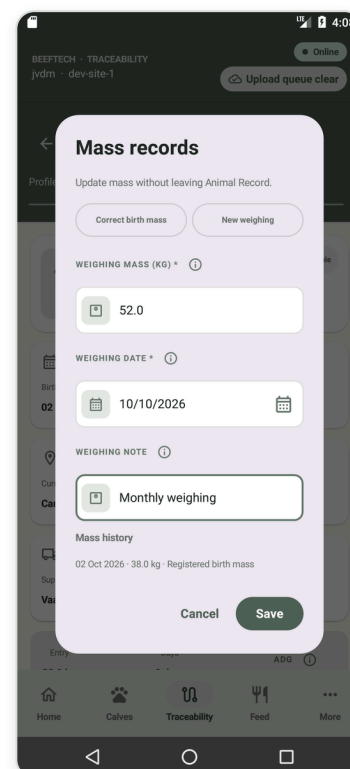
Find Animal searches by tag, number, breed, brand or parent (e.g. **B64**, **64**, **Bonsmara**). Filter by tag colour, or tap **Scan ear tag**.

Animal Record shows the animal's profile and status (for example *Moved*, *Sold* or *Deceased*), birth date, age, current location, registered mass, supplier, entry mass and **ADG** (average daily gain). Under **Linked records** you can open its movements, feed records, treatments, mortalities and costs.

Recording a weighing

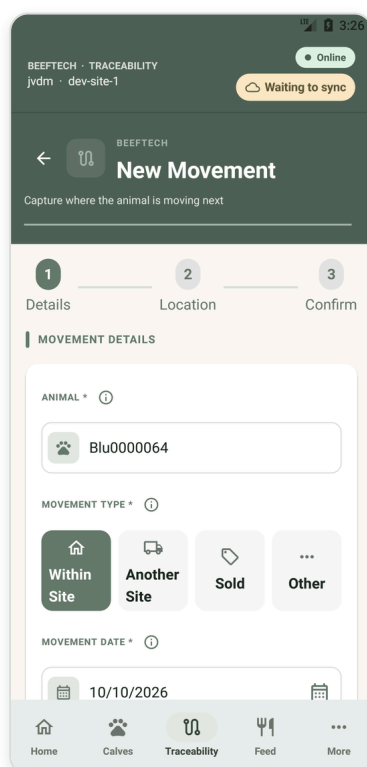
- 1 On the Animal Record, tap **Registered Mass**.
- 2 Choose **New weighing** to add a weighing, or **Correct birth mass** to fix the registered birth mass.
- 3 Enter the mass in kg (must be more than 0), the date and an optional note, then tap **Save**.

The **Mass history** list shows all earlier weighings. ADG is calculated once there are at least two masses on different dates.

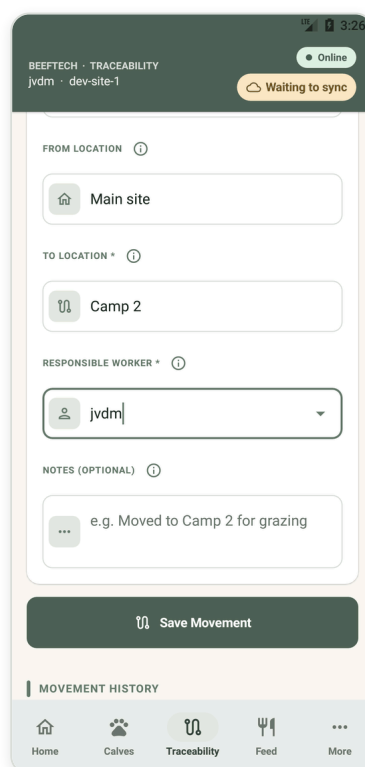


Mass records

Animal Movement



Movement type and date



Locations and responsible worker



- **Movement type ***: Within Site, Another Site, Sold or Other.
- **Movement date ***, **From location** and **To location *** (pick or type).
- **Responsible worker *** and optional **Notes**.

Tap **Save Movement**. The app will not save a movement to the place the animal already is. Past movements are listed under **Movement history**.

Supplier

Records where a purchased animal came from.

- **Supplier name ***: search registered suppliers or type a new one, then tap the *Use "..."* suggestion.
- **GLN number**: exactly 13 digits, or leave blank.
- **Date of purchase ***.
- **Purchase batch number**: created automatically (**PB-...**) once a supplier is chosen.

The **Batch overview** shows head in the batch and average entry mass. Tap **Save Supplier**.

The screenshot displays the 'Supplier' form in the BEEFTECH app. At the top, the header bar shows 'BEEFTECH · TRACEABILITY' and 'jvdm · dev-site-1'. Below the header, the title 'Supplier' is prominently displayed. The form contains several input fields: 'SUPPLIER NAME *' with a dropdown menu showing 'Vaal Weaners', 'GLN NUMBER' with a text input field containing '6009876543210', 'DATE OF PURCHASE *' with a date picker set to '10/10/2026', and 'PURCHASE BATCH NUMBER' with a text input field containing 'PB-20261010-VAA-B601'. The bottom of the screen features a navigation bar with icons for 'Home', 'Calves', 'Traceability', 'Feed', and 'More'.

Supplier



Location & Feed

- **Destination *** (pen or camp) and **Days in destination**.
- **Ration ***, **Days** on that ration and **Cost** in rand.
- Ration days cannot be more than the days at the destination.
- Enter the real cost. Zero is accepted only if the feed really cost nothing.

Tap **Save Location & Feed**. The feed cost also appears in the animal's [Costs](#).

Location & Feed

Treatments

Disease / condition

Treatment type

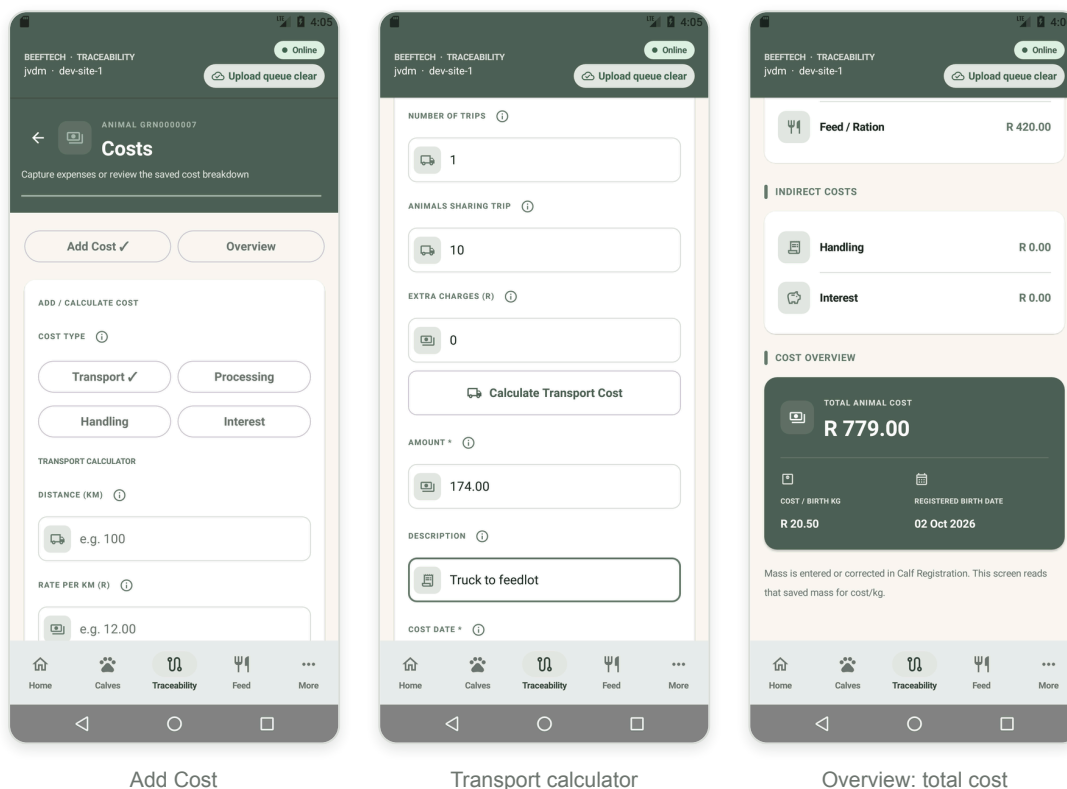
Batch, volume and cost

- 1 **Disease / condition ***: pick from the list or type a condition that is not listed.



- 2 **Treatment type** *: pick from the list (for example Antibiotic treatment, Vaccination, Deworming). Use **Other** if nothing fits.
- 3 The **Batch number** (TRT-...) is filled in automatically.
- 4 Enter the **Volume used** and the **Cost** in rand.
- 5 Tap **Save Treatment** . The app blocks an identical treatment from being saved twice.

Costs



The Costs screen has two tabs:

- **Add Cost** : choose the cost type — **Transport**, **Processing**, **Handling** or **Interest** — then enter the amount, an optional description and the date, and tap **Save Cost** .
 - **Transport calculator**: $\text{distance (km)} \times \text{rate per km} \times \text{number of trips} \div \text{animals sharing the trip}$, plus extra charges. Tap **Calculate Transport Cost** to fill in the amount.
 - **Interest calculator**: simple interest over a 365-day year.
- **Overview** : direct costs (Transport, Processing, Treatment, Feed / Ration), indirect costs (Handling, Interest, Other), the **total animal cost** and **cost per birth kg**.

NOTE

Treatment and Feed costs are added automatically from the Treatments and Location & Feed screens. Do not enter them again here.

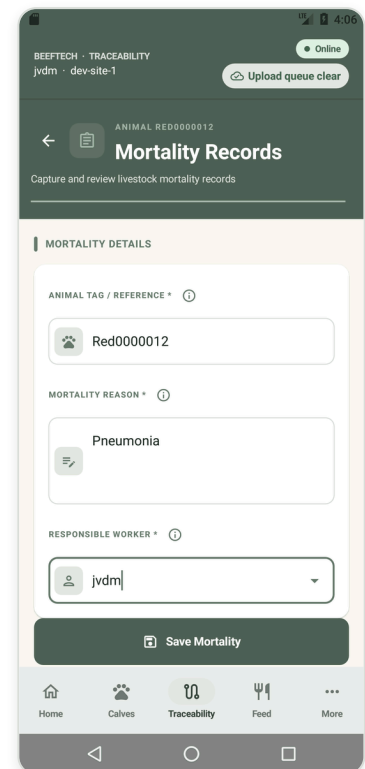


Mortality Records

- 1 Find the animal, then enter the **Mortality reason** * (e.g. illness, injury or unknown cause).
- 2 Enter the **Responsible worker** *.
- 3 Tap **Save Mortality**. The animal's status changes to **Deceased**.

CHECK FIRST

Make sure you have the right animal before saving. A mortality can only be undone by a manager voiding the record.

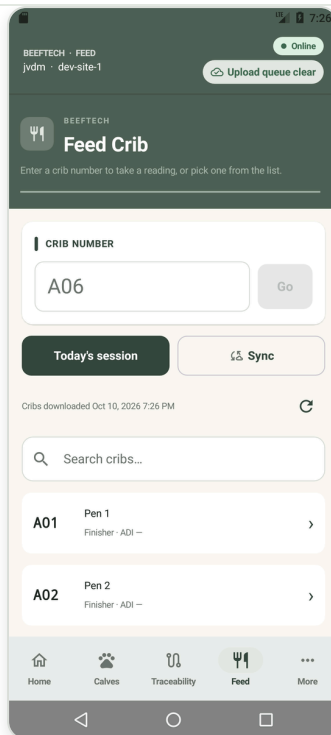


Mortality Records

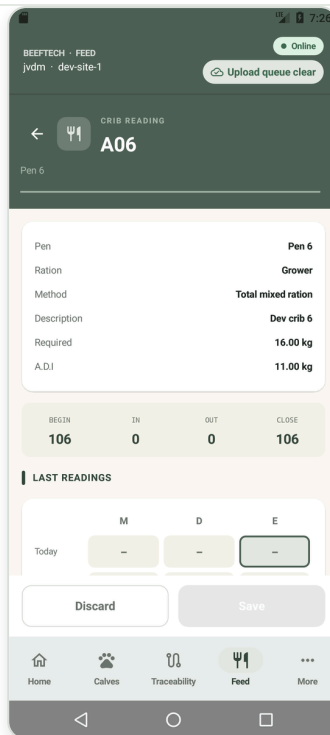
Feed tab

Use the **Feed** tab to take crib readings: how much feed is left in a crib, and the A.D.I (average daily intake) for its pen. Readings are saved on the phone, so you can work without a signal. They are sent to the server when the phone is online.

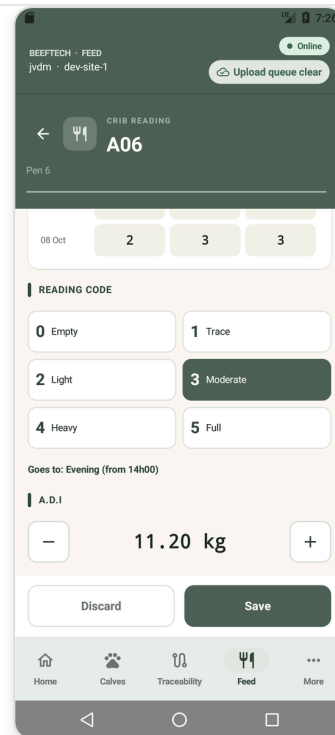
1. **Open a crib.** Type the crib number (for example **A06**) and tap **Go**, or tap a crib in the list. The list can be searched. When the tab opens the phone downloads your site's cribs; tap the refresh arrow to download them again.
2. **Check the details.** The crib shows its pen, ration, method, required feed and A.D.I, the animal counts (Begin, In, Out, Close) and a grid of the last three days. Rows are days; columns are **M** (Morning), **D** (Mid-Day) and **E** (Evening). The outlined block is where a reading saved now will go.
3. **Pick a reading code.** Tap one of the codes **0** to **5** (Empty to Full). Under the codes the app says where it will be filed, for example *Goes to: Evening (from 14h00)*. Readings before 11h00 go to Morning, before 14h00 to Mid-Day, and later ones to Evening.
4. **Adjust the A.D.I.** Tap **+** or **-**. Each tap changes it by 0.1 kg.
5. Tap **Save**. The app shows *Saved A06 · Evening 3* and returns to the crib number, ready for the next crib. Tap **Discard** to leave without saving; the app asks first if you changed something.



Crib number and list



Crib details



Code and A.D.I

NOTE

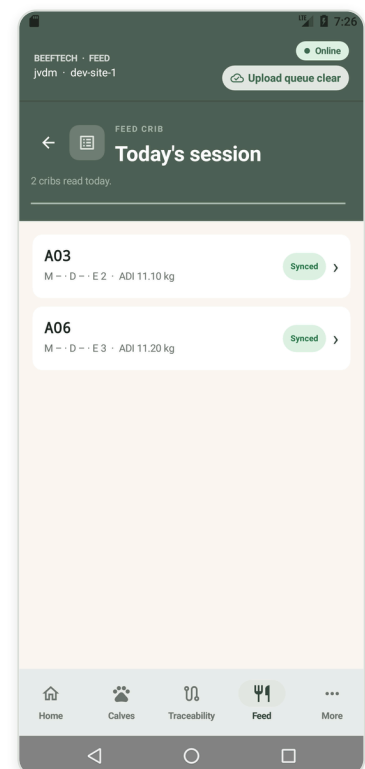
If you save a second reading in the same block on the same day, the newer code is the one shown in the grid. Earlier readings are kept.

Today's session. On the Feed tab, tap **Today's session** to see every crib you read today: its crib number, the codes for Morning, Mid-Day and Evening, and the A.D.I. A green **Synced** pill means the reading is on the server; **Waiting** means it will be sent when the phone is online. Tap a crib to see each reading with its time, block, code and A.D.I.

Sync. Tap **Sync** on the Feed tab to send waiting readings now. They are also included in **Sync now** on the Home screen. See [Working offline and syncing](#).

NO SITE, NO CRIBS

Cribs belong to a site. A user who has no site (an administrator, for example) cannot download cribs.



Today's session



My activity

More → **My activity** shows who is signed in, the role and site, how many of your records are waiting, and your sync history. See [Sync history](#).

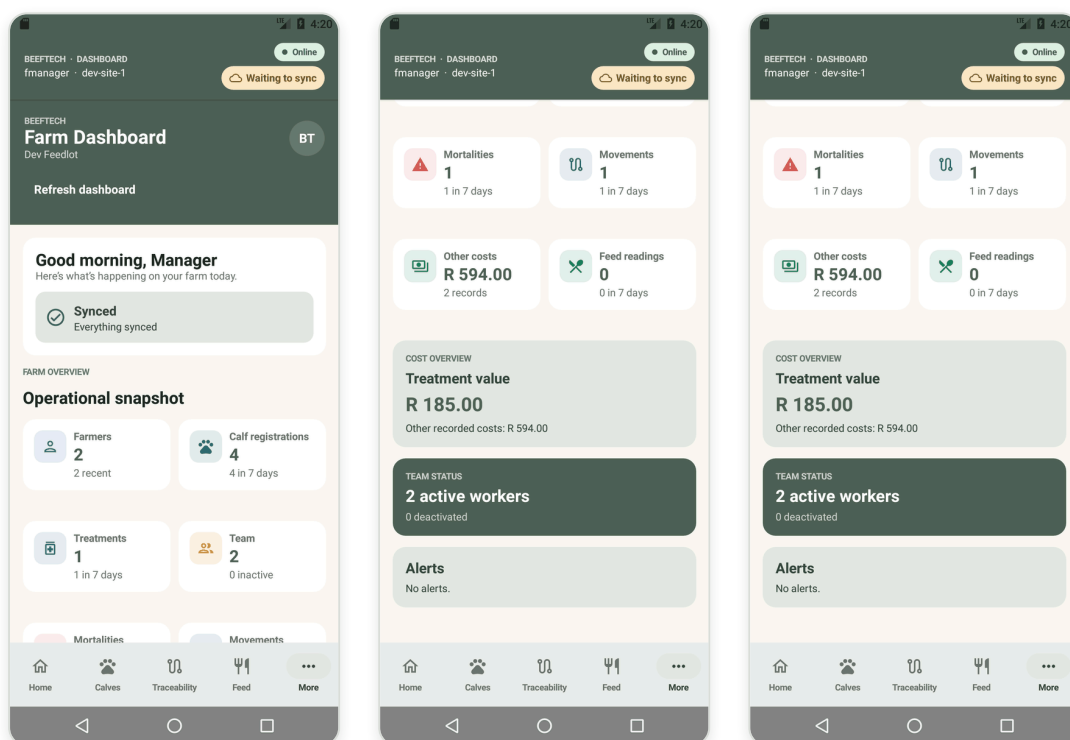
Site manager guide

Managers see the tools below under **More**, for their own site. These screens read live data from the server, so they need a connection. When offline they show “... needs a connection. Check your signal and tap Refresh.”

NOTE

If you are a manager and no site has been assigned to you, the Dashboard asks you to contact an administrator.

Dashboard



Farm overview

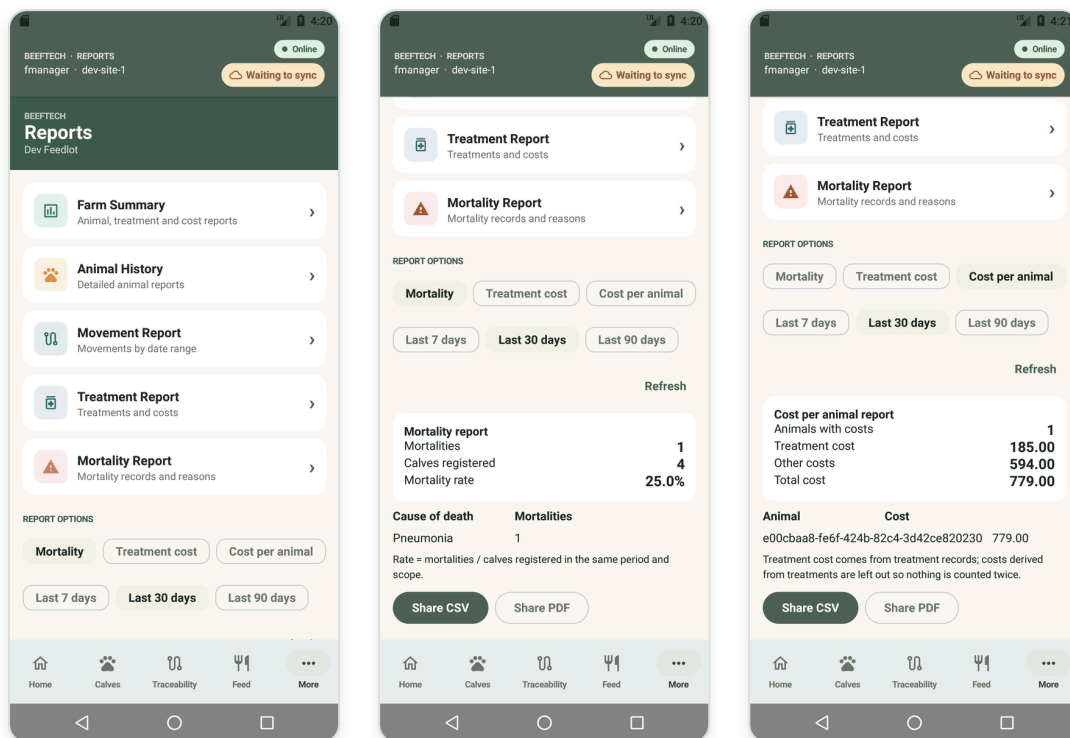
Costs and team status

Alerts

- **Operational snapshot:** farmers, calf registrations, treatments, team, mortalities, movements, other costs and feed readings, with the number in the last 7 days.
- **Cost overview:** treatment value and other recorded costs.
- **Team status:** active and deactivated workers.
- **Alerts:** for example a worker who has not connected for longer than the time set in [Sync policy](#).

Tap **Refresh dashboard** for the latest figures. Admins can switch between sites or choose **All sites**.

Reports



Report types and options

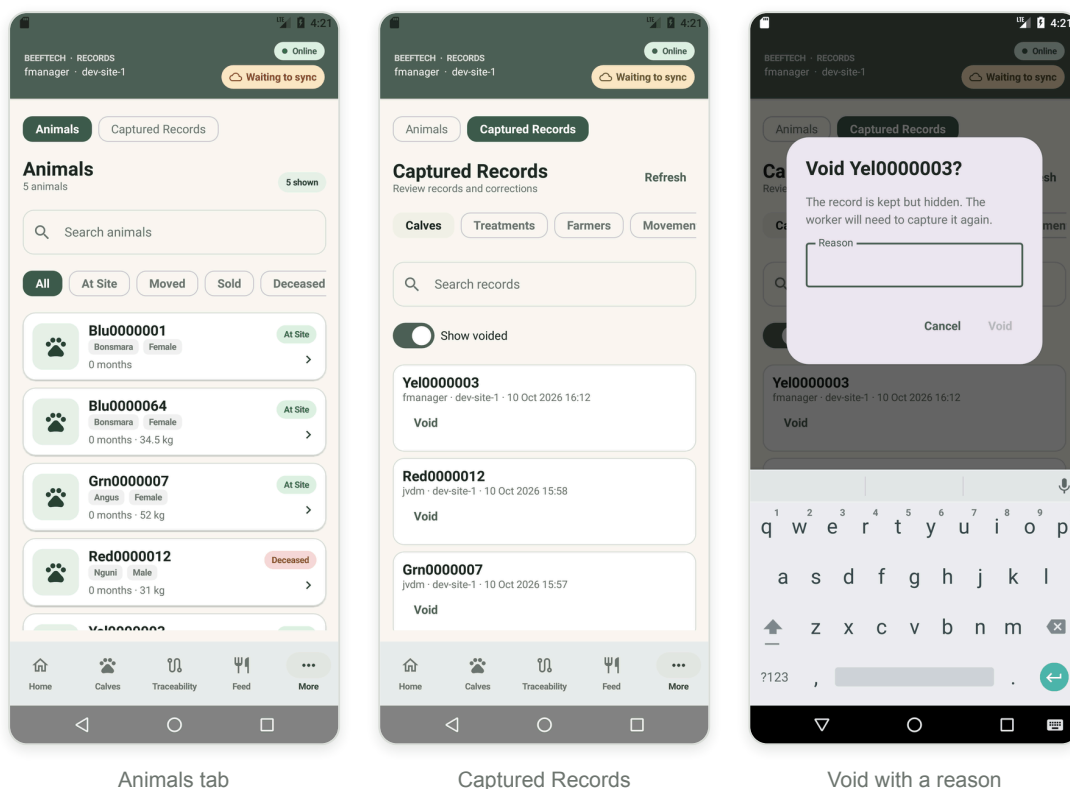
Mortality report

Cost per animal

- 1 Choose a report: **Farm Summary**, **Animal History**, **Movement Report**, **Treatment Report** or **Mortality Report**.
- 2 Under **Report options** choose the view (Mortality, Treatment cost, Cost per animal) and the period: **Last 7**, **30** or **90 days**.
- 3 Tap **Refresh** to build the report.
- 4 Tap **Share CSV** (for Excel) or **Share PDF** to send it by email, WhatsApp, Drive and so on.

Mortality rate = mortalities ÷ calves registered in the same period and site.

Records and voiding



Animals tab

Captured Records

Void with a reason

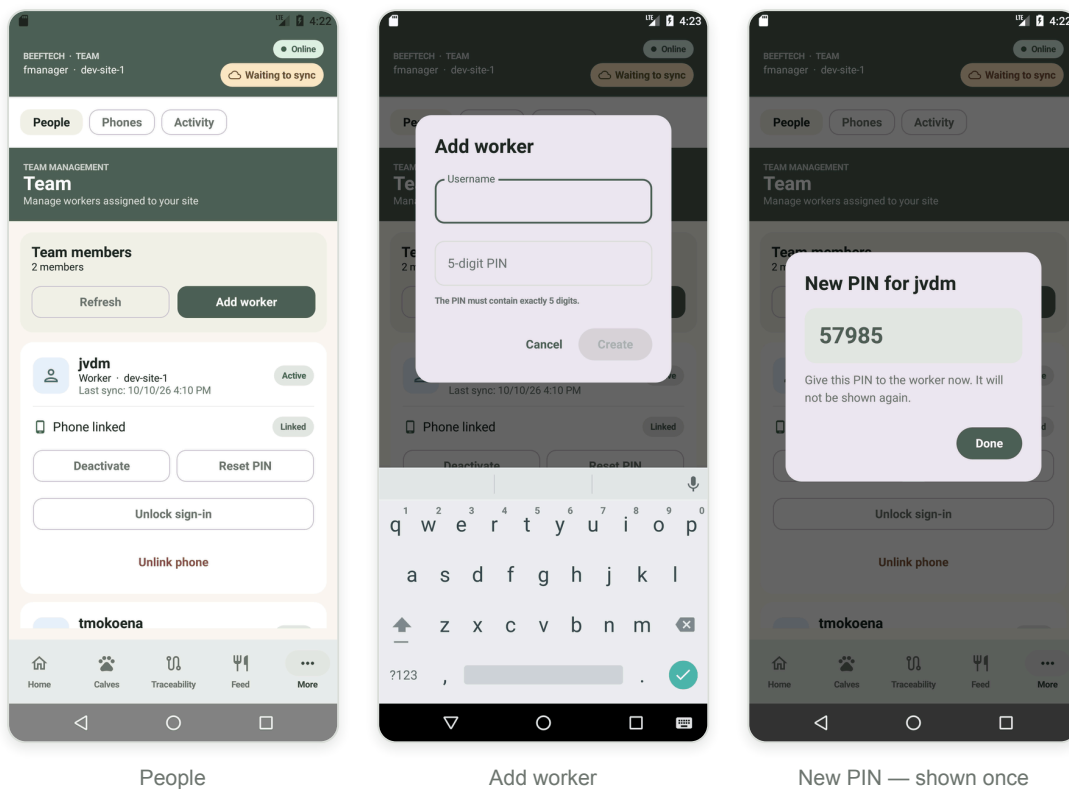
- **Animals:** every animal on the site, filtered by **At Site**, **Moved**, **Sold** or **Deceased**, with a search box.
- **Captured Records:** records by type (Calves, Treatments, Farmers, Movements, ...), showing who captured them and when.

Voiding a wrong record

- 1 Find the record under **Captured Records** and tap **Void**.
- 2 Type the **Reason** (required) and tap **Void**.

The record is kept but hidden from lists and reports, and the action is written to the audit log. The worker will need to capture the correct record again. Switch on **Show voided** to see voided records. Admins can also **assign a record to a site** (a reason is required).

Team



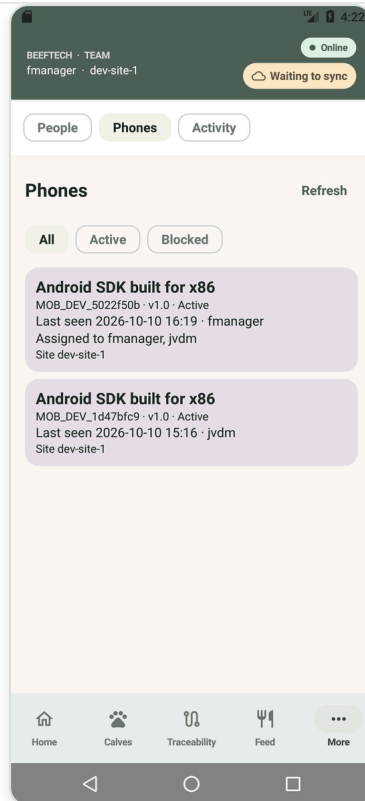
People

Add worker

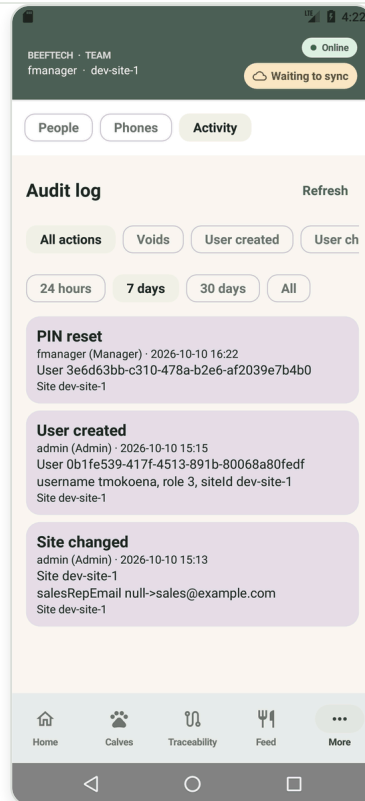
New PIN — shown once

Team has three sub-tabs: **People**, **Phones** and **Activity**. Managers see and manage the **workers on their own site**.

Action	What it does
Add worker	Create a worker with a username and a 5-digit PIN on your site. Give the worker the PIN in person.
Reset PIN	Immediately creates a new PIN and shows it once . Write it down and give it to the worker before tapping Done .
Unlock sign-in	Lifts the 5-minute lock after too many wrong PINs.
Unlink phone	Releases the worker from their current phone so they can sign in on a different one.
Deactivate / Reactivate	Stops a worker from signing in (for example when they leave), or lets them back in.



Phones: model, app version, last seen



Activity: audit log for the site

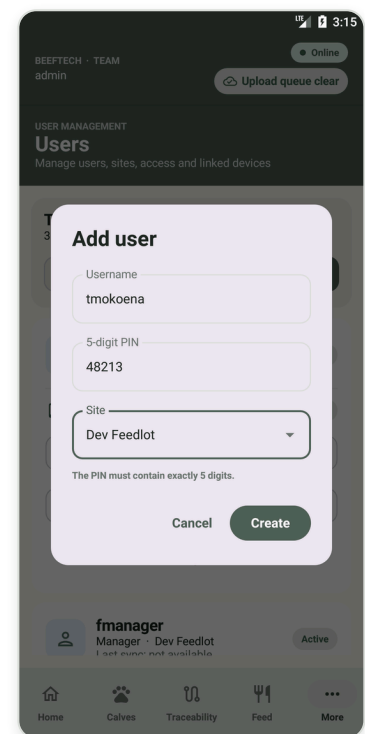
Administrator guide

Admins have every manager tool for all sites, plus **More** → **Admin**. The Admin area has tabs along the top — swipe the tab row sideways to reach them all: Sites, Phones, Login security, Reference data, Sync policy, Sync security and Audit log.

Users across all sites

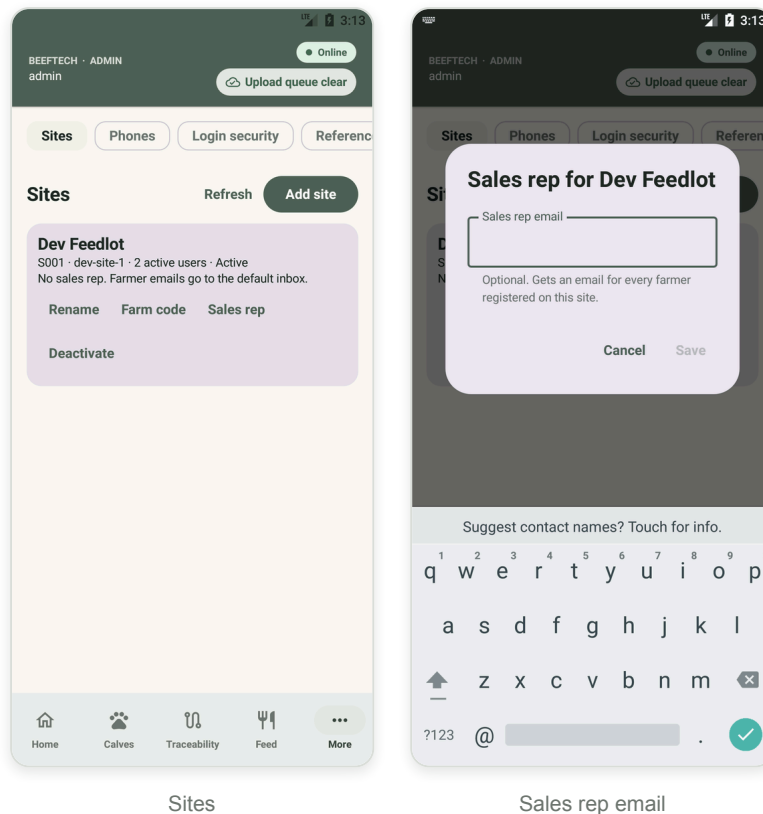
For admins, **More** → **Team** becomes **User management** for every user on every site.

- **Add user**: username, 5-digit PIN and site.
- Reset PIN, unlock sign-in, unlink phone, deactivate/reactivate — as for managers.
- **Change site** for a user. You cannot change your own site or another admin's.



Add user

Sites



Sites

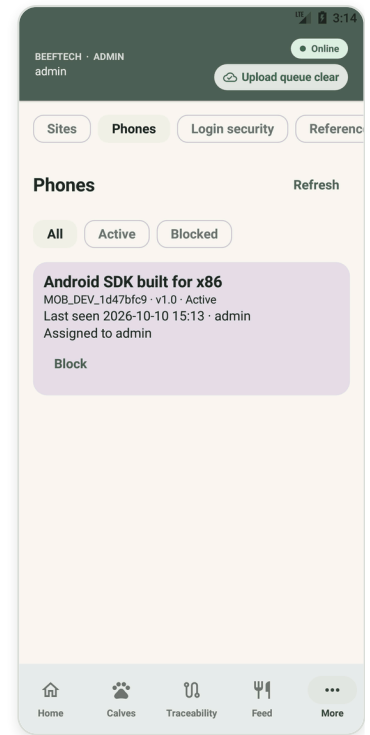
Sales rep email

- **Add site** and **Rename**.
- **Farm code**: a 4-character code for the site (capital letters A–Z and digits 0–9), used in the names of sync batches, photos and exported reports.
- **Sales rep**: an optional email address. That person receives an email for every farmer registered on the site. Without one, farmer emails go to the default inbox.
- **Deactivate** a site that is no longer used.

Phones

Lists every phone that has signed in: model, phone ID, app version, last seen, which users it is assigned to and its site. Filter by **All**, **Active** or **Blocked**.

Tap **Block** to stop a lost or stolen phone from signing in. Its users then see *“This phone has been blocked. Contact your administrator.”* Unblock it the same way.

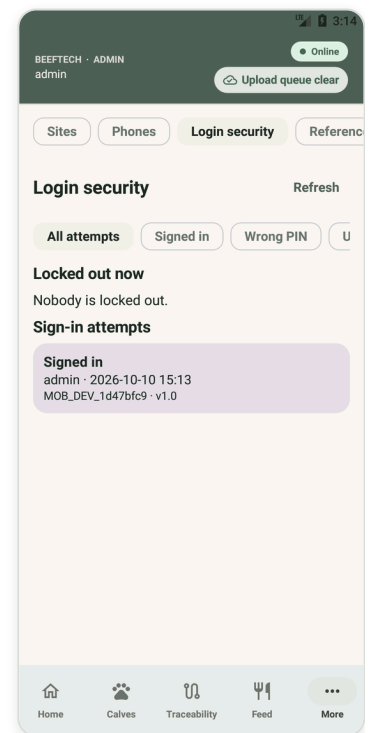


Phones

Login security

Shows who is **locked out now** (with an unlock action) and a list of **sign-in attempts** — successful sign-ins, wrong PINs and other outcomes — with the user, time and phone.

Use it to spot repeated failed attempts on an account.



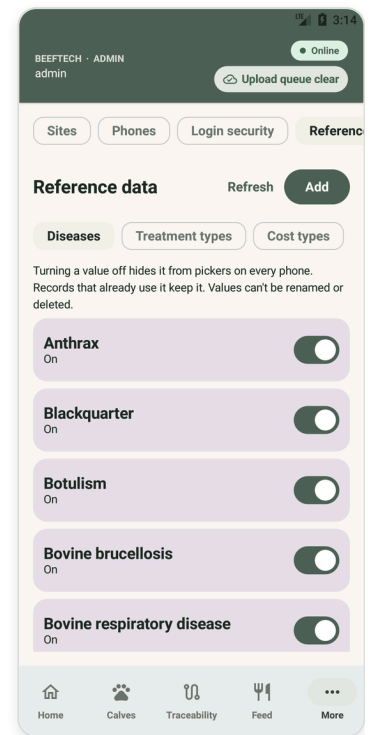
Login security

Reference data

Manages the pick lists on every phone: **Diseases**, **Treatment types** and **Cost types**.

- Switch a value **off** to hide it from the pickers on every phone. Records that already use it keep it.
- **Add** a new value with a code such as `VET_CALLOUT` and a display name.
- Values cannot be renamed or deleted — switch off the old one and add a new one.

Phones receive changes the next time they connect.

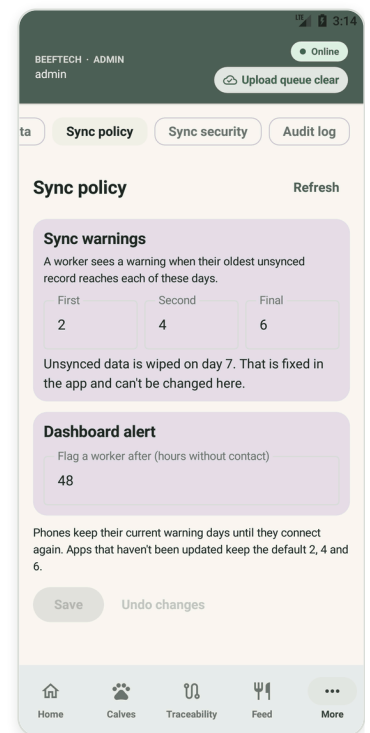


Reference data

Sync policy

- **Sync warnings:** the three days on which workers see the unsynced-data warnings (default 2, 4 and 6). Each must be between 1 and 6 and later than the one before.
- Unsynced data is wiped on **day 7**. This is fixed in the app and cannot be changed here.
- **Dashboard alert:** flag a worker on the Dashboard after this many hours without contact (default 48).

Tap **Save**. Phones keep their current warning days until they next connect.

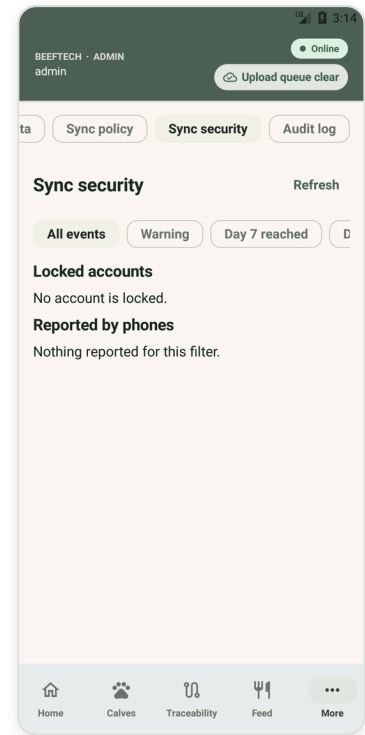


Sync policy

Sync security

Lists **locked accounts** — accounts locked by the 7-day rule — and the warning and day-7 events that phones have reported.

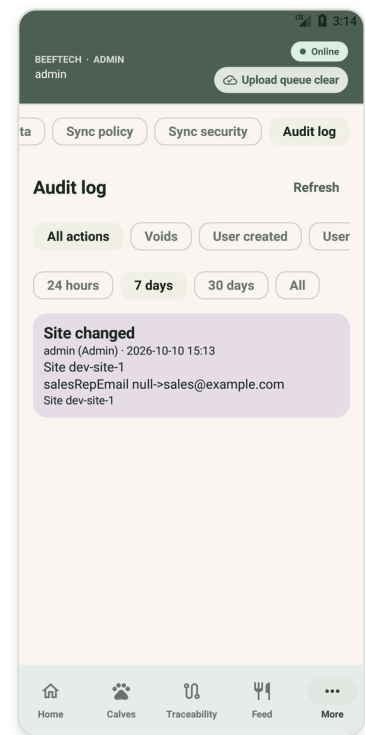
To restore a worker after a day-7 lock, find the account under **Locked accounts** and clear the lock. Then ask the worker to tap **Check again** on their phone.



Sync security

Audit log

Every administrative change is recorded: users created and changed, PIN resets, voids, site changes and more. Filter by action type and by period (24 hours, 7 days, 30 days, All).



Audit log

IT and installer guide

This chapter is for the person who installs the app and prepares phones. Server installation and configuration are covered in the project `README.md`.

Installing the app

Requirement	Detail
Phone	Android 7.0 (API 24) or later, with a rear camera (tag scanning) and GPS (farm location).
App file	The BeefTech APK supplied by the development team (package <code>com.beeftech.demoapp</code> , version 1.0).
Permissions	Camera (scanning tags and calf photos) and Location (farm coordinates). The app asks the first time each is needed.

- 1 Copy the APK to the phone, or download it from the link you were given.
- 2 Open it and allow installation from this source if Android asks.
- 3 Tap **Install**, then **Open**.

Developers can install directly from the source code with `./gradlew :demoapp:installDebug`.

Server address

The server address is built into the app when it is compiled. It cannot be changed on the phone.

Build	Server
Release	<code>https://beeftech-backend.onrender.com/</code>
Debug (emulator)	<code>http://10.0.2.2:8081/</code> — the developer's computer, as seen from the Android emulator.
Debug (real phone)	Build with <code>-Pbeeftech.baseUrl=http://<server-ip>:8081/</code> , e.g. <code>./gradlew :demoapp:installDebug -Pbeeftech.baseUrl=http://192.168.1.20:8081/</code>

Setting up a new phone

- 1 An admin or manager creates the user's account ([Team](#) or [User management](#)) and gives them their username and PIN.
 - 2 Install the app and make sure the phone has a **working internet connection**.
 - 3 The user signs in once online. The account is now linked to this phone, and the phone appears under [Phones](#).
- Let the phone sync once (tap **Sync now**) so it downloads the pick lists and the sync policy.



- 5 Optionally, in **Assign Animal to Farmer** → **Tools & sync**, download the farm's animals for offline use.

Replacing or losing a phone

- **New phone for an existing user:** on the old phone, sync until **Upload queue clear**. Then, in Team, tap **Unlink phone** for the user and sign in on the new phone while online.
- **Lost or stolen phone:** in Admin → Phones, tap **Block**, then unlink the user and set them up on a new phone. Records on the lost phone that never synced cannot be recovered.
- **Clearing app data or reinstalling** removes unsynced records and gives the phone a new identity. The user must be unlinked from the old identity before signing in again.

Before going live

SECURITY CHECKLIST

- The current build protects the phone database with a fixed demonstration key. Replace it with a key kept in the Android Keystore before production use.
- Remove or change the seeded test accounts (`admin`, `fmanager`, `jvdm`) on the production server.
- Use the HTTPS server address for release builds.
- Set the app name and launcher icon to BeefTech branding.



Troubleshooting and glossary

Messages and what to do

Message	Meaning	What to do
Incorrect username or PIN	The username or PIN is wrong.	Check spelling and try again. After five tries you are blocked for a while.
Too many attempts. Try again in 1 minutes.	Five wrong PINs on this phone.	Wait one minute. If it continues online, ask a manager to Unlock sign-in .
This phone is registered to another worker	Your account is linked to a different phone.	Use your own phone, or ask a manager to Unlink phone .
This phone has been blocked. Contact your administrator.	An admin blocked the phone.	Contact your administrator.
Your access was ended. Connect to the internet and sign in again.	Your account was deactivated or changed.	Connect and sign in. If it still fails, contact your manager.
Connect once to set up this account	First sign-in on this phone, but there is no signal.	Find signal and sign in online once.
Cannot reach the server. Try again.	No connection to the server.	Check signal or Wi-Fi and try again.
Your session has ended. Sign in again.	Your session expired.	Sign in again. Your saved records are kept.
Data Not Synced / Critical Sync Warning / Final Sync Warning	Records have been waiting for 2 / 4 / 6 days.	Connect and tap Sync now — see The 7-day rule .
Account locked	Day 7 reached with unsynced data.	Contact an administrator to clear the lock, then tap Check again .
Unable to verify account security	The app could not check the account status.	Tap Try again when you have signal.
... needs a connection. Check your signal and tap Refresh.	A management screen needs live data.	Find signal and tap Refresh .
Client Code already exists!	Another farmer has that client code.	Use a different, unique client code.
Calf shows NEEDS ATTENTION	The server rejected it, usually a duplicate tag.	Open the calf, read the server message, then fix it or tap Retry sync .



Frequently asked questions

Can I work all day without signal?

Yes. Everything is saved on the phone. Sync when you get signal, and at least once every couple of days — see [The 7-day rule](#).

Two of us share a phone. Is that a problem?

Each person should sign in with their own account and log out when handing over the phone, so records are captured under the right name. Sync before handing over.

I registered a calf with the wrong tag. How do I fix it?

Calf details cannot be edited after saving. Register the calf again with the correct tag and ask your manager to void the wrong record with a reason.

Why can't I see a farmer or animal that a colleague registered?

Records reach other phones through the server. Make sure both phones have synced. For animals, use

Tools & sync → **Download farm animals for offline use**.

The scheduled sync didn't run at exactly 05:00.

Android may delay background work to save battery. The sync runs as soon as Android allows and there is a connection. Tap **Sync now** if you need it immediately.

Glossary

Term	Meaning
ADG	Average daily gain — the animal's weight gain per day between weighings.
Batch number	An automatic reference: PB-... for a purchase batch (Supplier), TRT-... for a treatment.
Dam / Sire	The calf's mother / father.
Ear tag ID	Colour code plus seven digits: BLu , Red , Grn or YeL , e.g. BLu0000064 .
FA Code (RMIS)	The farm's code in the Red Meat Industry Services system.
Farm code	A 4-character code for a site, set by an admin.
GLN	Global Location Number — a 13-digit number identifying a farm or supplier location.
PENDING	Saved on the phone, not yet confirmed by the server.
SYNCED	Confirmed by the server.
NEEDS ATTENTION	The server refused the record. Open it to see why.
Site	A feedlot or farm operation. Every user and record belongs to a site.
Sync	Sending records from the phone to the server and receiving updates back.
Void	Hiding a wrong record without deleting it. Voids are recorded in the audit log.